# Midterm 2

Prof. Jordan C. Hanson

April 23, 2025

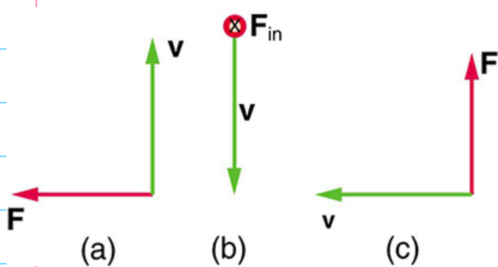



Figure 1: Several geometries for the Lorentz force on a charged particle moving in a B-field.

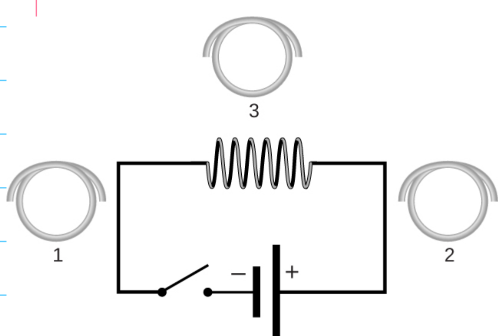



Figure 2: A battery provides and emf to a simple circuit with a solenoid, surrounded by three wire loops *in the plane of the circuit.*

## 1 Unit 4: Magnetism II

1. Determine the direction of the B-field in Fig. 1 (a)-(c), given a positively charged particle moving with velocity $\vec{v}$.

   A) Into the page

   B) To the right

   C) Out of the page

2. Consider Fig. 2. (a) What is the direction of the current induced in coil 1 when the switch is closed? If it is left closed for a long time? When it is opened? (b) Repeat exercise (a) for coil 2. (c) Repeat exercise (a) for coil 3.

| Switch | Coil 1 | Coil 2 | Coil 3 |
|---|---|---|---|
| closed | counterclockwise | counterclockwise | counterclockwise |
| left closed | No change | No change | No change |
| open | Clockwise | Clockwise | Clockwise |

3. (a) An MRI technician moves his hand from a region of very low magnetic field strength into an MRI scanner's 2.00 T field with his fingers pointing in the direction of the field. Find the average emf induced in his wedding ring, given its diameter is 2.20 cm and assuming it takes 0.250 s to move it into the field. (b) What current is induced in the ring if its resistance is 0.0100 Ω? (c) Calculate the average power dissipated in the ring.

   A) $3.04 \times 10^{-3}$ V

   B) 0.304 A

   C) $9.24 \times 10^{-4}$ W

4. (a) A bicycle generator rotates at 1875 rad/s, producing an 18.0 V peak emf. It has a 1.00 by 3.00 cm rectangular coil in a 0.640 T field. How many turns are in the coil? (b) What is the period of the AC output of the generator?

   A) 50 turns

   B) $3.35 \times 10^{-3}$ seconds

5. An American traveler in New Zealand carries a transformer to convert New Zealand's standard 240 V to 120 V so that she can use some small appliances on her trip. (a) What is the ratio of turns in the primary and secondary coils of her transformer? (b) What is the ratio of input to output current?

   A) 2/1

   B) 1/2

# 1 Unit 4: Magnetism II

1) A) Into the page

B) To the right

C) Out of the page

2a)

| Switch | Coil 1 | Coil 2 | Coil 3 |
|---|---|---|---|
| Closed | counterclockwise | counterclockwise | counterclockwise |
| left closed | No change | No change | No change |
| Open | Clockwise | Clockwise | Clockwise |

3a) 2.00T field
ring = 2.20 cm
0.250s to move

$A = \pi r^2 \quad \frac{0.0220}{2} = 0.0110m$

$A = \pi (0.0110)^2 = 3.8 \times 10^{-4} m^2$

$emf_{avg} = \frac{\Delta \Phi}{\Delta t} \quad emf_{avg} = \frac{7.60 \times 10^{-4}}{0.250} = \underline{3.04 \times 10^{-3} V}$

$\Delta \Phi = B \times A = 2 \times 3.8 \times 10^{-4} = 7.6 \times 10^{-4} Wb$

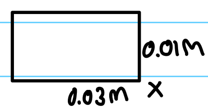
b) $I = \frac{emf}{R} = \frac{3.04 \times 10^{-3}}{0.0100} = \underline{0.304 A}$

c) $P = I^2 R$

$P = (0.304)^2 \times 0.0100 = \underline{9.24 \times 10^{-4} W}$

4a) 1875 rad/s $\longrightarrow$ 18.0V

1x3 cm coil in 0.640 T field




$A = 3 \times 10^{-4}$

$V_p = N \cdot A \cdot B \cdot \omega$

$N = \frac{V_p}{A \cdot B \cdot \omega} \longrightarrow N = \frac{18}{3 \times 10^{-4} (0.64)(1875)}$

$\underline{N = 50 \text{ turns}}$

b) $\omega = 2\pi f \longrightarrow f = \omega / 2\pi$

$\frac{1875}{2\pi} = 298.4 Hz$

$T = 1/298.4 Hz = \underline{3.35 \times 10^{-3} \text{ seconds}}$

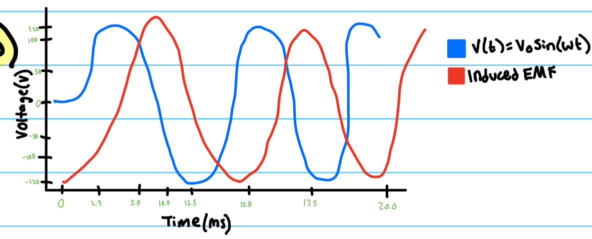
5a) 240V to 120V

$\frac{240}{120} = \underline{\frac{2}{1}}$

b) $\frac{120}{240} = \underline{\frac{1}{2}}$

6. Consider Fig. 2. Suppose coil 3 ($N = 1$ turn) is rotated 90 degrees such that its normal direction is aligned with the normal direction of the solenoid ($N = 6$ turns). The system is now acting as a **transformer**. Suppose the emf in the solenoid is given by $V(t) = V_0 \sin(\omega t)$. (a) Draw a graph of $V(t)$, with axes labeled. (b) Add the induced emf in coil 3 to your graph, assuming the magnetic flux in coil 3 is rendered identical to that in the solenoid with an iron core. (c) If $V_0 = 120$ V, and $\omega = (240\pi)$ Hz, when does $V(t) = 0$?

A+B)




C) 1/120 seconds

7. Camera flashes charge a capacitor to high voltage by switching the current through an inductor on and off rapidly. In what time must the 0.100 A current through a 2.00 mH inductor be switched on or off to induce a 500 V emf?

$4\times10^{-7}$ s

8. A large research solenoid has an inductance of 25.0 H. (a) What induced emf opposes shutting it off when 100 A of current through it is switched off in 80.0 ms? (b) How much energy is stored in the inductor at full current? (c) At what rate in watts must energy be dissipated to switch the current off in 80.0 ms? (d) Design a solenoid that has an inductance of 25.0 H by choosing the number of turns $N$, the cross-sectional area $A$, and the number of turns $l$. (d) What is the reactance of the inductor at 10 kHz?

A) 31,250V or 31.25kV

B) 125,000J or 125kJ

C) 1,562,000W or 1.562MW

D) $A = 0.01m^2$, $l = 0.5m$, $N = 31{,}539.15$ turns

e) $1.56\times10^{6}\,\Omega$

9. Your RL circuit has a characteristic time constant of 20.0 ns, and a resistance of 5.00 MΩ. (a) What is the inductance of the circuit? (b) What resistance would give you a 1.00 ns time constant, perhaps needed for quick response in an oscilloscope? (c) What percentage of the final current $I_0$ flows 3.0 ns after deactivation?

A) 0.1 H

B) $1\times10^{8}\,\Omega$

C) 86.1%

10. Consider Fig. 3. Assume the bottom wire is grounded at 0V. (a) Using Kirchhoff's loop rule on the left side of the circuit, write an equation that describes changes in potential. (b) Repeat exercise (a) for the right loop. (c) Solve parts (a) and (b) for $v_{\text{in}}$ and $v_{\text{out}}$, respectively, and calculate $v_{\text{out}}/v_{\text{in}}$ in terms of $R$ and $C$. (d) What is the value of $v_{\text{out}}/v_{\text{in}}$ for $f = 100(2\pi RC)^{-1}$? For $f = 0.1(2\pi RC)^{-1}$?

A) $v_{in} = iR$

B) $V_{out} = v_C$

C) $\frac{V_{out}}{V_{in}} = \frac{1}{1+j\omega RC}$

D) At $f = 100(2\pi RC)^{-1} = 0.01$
At $f = 0.1(2\pi RC)^{-1} = 0.995$

11. (a) What is the resonance frequency, $f_0$, of an RLC circuit with $R = 0.1$ kΩ, $C = 1$ μF, and $L = 10$ mH? (b) What is the total impedance of this circuit at $f = 0.1f_0$ and $f = 10f_0$?

A) 1,592.36 Hz

B) $0.1f_0 = 995.03$ Hz and $10f = 995.03$ Hz

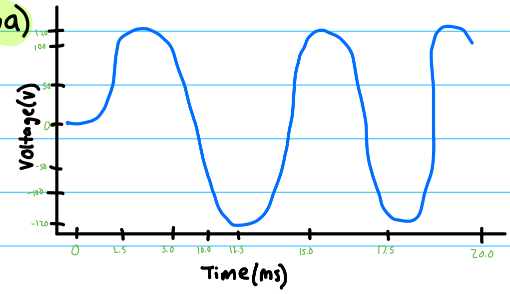
12. For an RLC circuit with that values of $R$, $L$, and $C$ from the previous problem, an AC source voltage characterized by $V_{\text{rms}} = 120$ V, (a) what is $I_{\text{rms}}$ at $10f_0$ and $0.1f_0$? (b) What is $P_{\text{rms}}$ at these frequencies? *Hint: calculate the phase angle between resistance and reactance.*

A) 0.1206 A

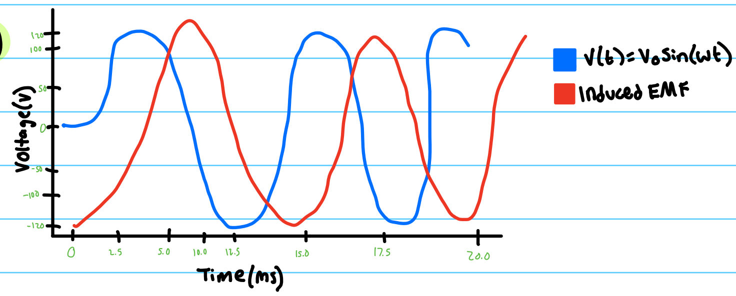
B) 1.45W

13. Suppose we are using an LC resonator and diode combination to create an AM radio signal. If the

6a)




b)




c) $V_0=120V \quad \omega=(240\pi)Hz \quad V(t)=0$

$V(t)=120\sin(240\pi t)$

$t=\frac{1}{120}$ seconds

7) 0.100 A, 2.00mH, 500V

$Emf=L\frac{\Delta I}{\Delta t} \rightarrow \Delta t=L\frac{\Delta I}{Emf}$

$\Delta t=\frac{2\times10^{-3}(0.1)}{500}=4\times10^{-7}s$

8a) $25.0H=L$

$EMF=-L\frac{\Delta I}{\Delta t}=-25\times\frac{0-100}{80\times10^{-3}}$

$EMF=31,250V$ or $31.25kV$

b) $W=\frac{1}{2}LI^2$

$\frac{1}{2}\cdot25(100^2)=125,000J$ or $125kJ$

c) $P=\frac{W}{t}=\frac{125\times10^{3}}{80\times10^{-3}}=1,562,000W$ or $1.56mW$

d) $L=25H \quad A=0.01m^2 \quad l=0.5m \quad N=?$

$L=\frac{\mu_0N^2A}{l}$

$25=\frac{(4\pi\times10^{-7})N^2(0.01m^2)}{0.5m}=N^2=\frac{25\times0.5}{(4\pi\times10^{-7})0.01}=\sqrt{994718394.3}=31,539.15$

e) $X_L=2\pi fL=2\pi\cdot10^4\cdot25=1.57\times10^6\Omega$

9a) 20.0ns, 5.00MΩ

$\tau=\frac{L}{R}=L=R\cdot\tau$

$(5\times10^6)20\times10^{-9}=0.1H$

b) $R=\frac{L}{\tau}\rightarrow\frac{0.1}{1\times10^{-9}}=1\times10^8\Omega$

c) $I(t)=I_0e^{-t/\tau}\rightarrow e^{-t/\tau}=\frac{I(t)}{I_0}=e^{-3/20}=e^{-0.15}=0.861$

$\downarrow$

86.1%

10a) $v_{in}=iR$

b) $v_{out}=v_C$

c) $\frac{V_{out}}{V_{in}}=\frac{1}{1+j\omega RC}$

d) $\omega=2\pi f=2\pi\cdot100\cdot\frac{1}{2\pi RC}=\frac{100}{RC}$

$\frac{V_{out}}{V_{in}}=\frac{1}{\sqrt{1+100^2}}=0.00999995\rightarrow0.01$

$\frac{V_{out}}{V_{in}}=\frac{1}{\sqrt{1+0.1^2}}=0.995$

11a) $R=0.1k\Omega$, $C=1\mu F$, $L=10mH$
$1\times10^{-6}$ $10\times10^{-3}$

$f_0=\frac{1}{2\pi\sqrt{LC}}$

$f_0=\frac{1}{2\pi\sqrt{10\times10^{-3}H\times1\times10^{-6}F}}=1,592.36Hz$

b) $Z=\sqrt{R^2+(2\pi fL-\frac{1}{2\pi fC})^2}$

$f=0.1f_0\times1592.36=159.236Hz$

$\sqrt{(0.1\times10^3)^2+(2\pi\times159.236\times10\times10^{-3}-\frac{1}{2\pi\times159.236\times10^{-6}})^2}$

$=995.03Hz$

$f=10f_0=10\times1,592.36=15,923.6Hz$

$\sqrt{(0.1\times10^3)^2+(2\pi\times15,923.6\times10\times10^{-3}-\frac{1}{2\pi\times15,923.6\times10^{-6}})^2}$

$=995.03Hz$

12a) $V_{rms}=120V$ $R=0.1k\Omega$, $C=1\mu F$, $L=10mH$
$100\Omega$ $1\times10^{-6}$ $10\times10^{-3}$

$I_{rms}=\frac{V_{rms}}{|Z|}=\frac{120}{995}=0.1206A$
or
120.6mA

b) $P_{rms}=V_{rms}\cdot I_{rms}\cdot\cos\theta$

$\theta=\frac{R}{|Z|}=\frac{100}{995}=0.1005$

$P_{rms}=120\cdot0.1206\cdot0.1005=1.45W$

13) $f_c=1.4MHz$

$f_c+f_m=1.4MHz+10kHz=1.410MHz$

$f_c-f_m=1.4MHz-10kHz=1.390MHz$

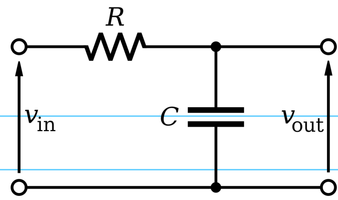



Figure 3: This RC circuit can act as a low-pass filter.

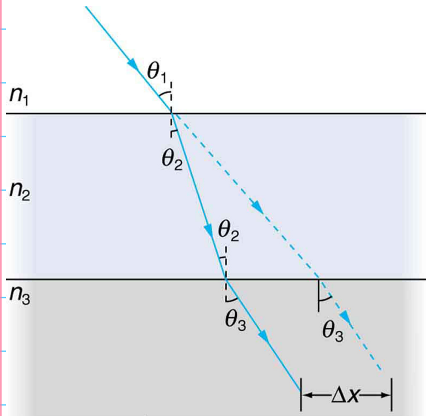



Figure 4: A light ray is in a medium with $n_1$, then enters a medium with $n_2$, then a medium with $n_3$.

resonance frequency of our resonator is 1.4 MHz, and our audio signal is at 10 kHz, what three frequencies should be present in our radio spectrum?

$f_1$ = 1.390 MHz or $1.39 \times 10^6$ Hz Lower sideband

$f_2$ = 1.400 MHz or $1.4 \times 10^6$ Hz

$f_3$ = 1.410 MHz or $1.41 \times 10^6$ Hz upper sideband

## 2 Unit 5: Waves, Optics, Medical Physics

1. (a) What is the B-field generated 1 cm laterally from a capacitor in an RC circuit that charges from 0 to 200 nC in 2 $\mu$s? (b) What *current* is responsible for this B-field, since the capacitor is not passing DC current like a wire?

   A) $2.0 \times 10^{-6}$ T = 2.0 $\mu$T

   B) 0.1A is responsible b/c it can still create a magnetic current due to the changing plates

2. (a) An aircraft receives a pulse reflected from glacial ice below it 10 $\mu$s after transmission. What is the displacement between aircraft and ice? (b) What fraction of the power is reflected from the ice, assuming standard indexes of refraction for air and ice at RF wavelengths?

   A) 1500m

   B) 7.87% Power reflected

3. (a) A microwave generates 1 kW of power, projected onto a 10 cm-squared area at a distance of 1 m. What is the intensity (power per unit area)? (b) If the intensity drops as the distance squared, what is the intensity 2 meters from the microwave?

   A) $1.0 \times 10^6$ W/m$^2$

   B) $2.5 \times 10^5$ W/m$^2$

4. (a) Show[1] that $\theta_3 = \theta_1$ in Fig. 4, if $n_1 = n_3$. (b) A doctor examines the ear of a child using a 12.0 cm focal length eyepiece 8.0 cm from the ear. What is the magnification?

   A) If $n_1 = n_3$, $\sin\theta = \sin\theta_3$ so $\theta_1 = \theta_3$

   B) 3.0

5. Suppose x-rays have a cross section of 200 barns at 10 keV in a lead vest protecting a medical patient. If the vest is 1 cm thick, what fraction of the x-rays pass through it?

   so 0.137% of Xrays pass through

6. (a) What is the radioactive dose in rads if 250 mJ of total energy is deposited in the body of a 60 kg adult? (b) What instead would be the dose if the radiation was concentrated in just 2.0 kg of tissue? (c) Assuming the RBE is 1 (x-rays or $\gamma$ rays), what is the dose from part (b) in Sv? (d) Does this dose represent a serious health risk?

   A) 0.4167 or 0.417 rad

   b) 12.5 rad

   c) 0.125 Sv

   d) This dosage is not an immediate risk but it can cause longlasting effects as the limit is ~2-3 mSV/year.

[1] *This proof is the reason why the center of our lenses (like those in our glasses) do not change ray directions. If our glasses changed the apparent location of objects, they would be useless.*

1a) $B=\frac{\mu_0 I_d}{2\pi r}$   $I_d=\frac{\Delta Q}{\Delta t}$

$r=1cm$   $\Delta Q = 200nC \rightarrow 2\times10^{-7}C$

$\Delta t = 2\mu s \rightarrow 2\times10^{-6}s$

$B=\frac{(4\pi\times10^{-7})(0.1)}{2\pi(1\times10^{-2})}=\underline{2.0\times10^{-6}T}$

b) $I_d=\frac{\Delta Q}{\Delta t}=\frac{2\times10^{-7}}{2\times10^{-6}}=\underline{0.1\ A}$

2a) $D=c\cdot\Delta t$

$c=3\times10^8 m/s$   $D=\frac{(3\times10^8)\cdot10\times10^{-6}}{2}=\underline{1500m}$

$\Delta t=10\mu s$

b) $R=\left(\frac{n_1-n_2}{n_1+n_2}\right)^2$

Air $\approx 1\ (n_1)$   $R=\left(\frac{1-1.78}{1+1.78}\right)^2=0.280^2=0.0784$

Ice $\approx 1.78\ (n_2)$

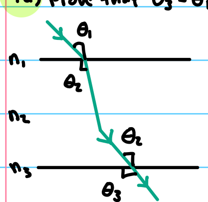
<u>7.87% of power is reflected</u>

3a) $I=\frac{P}{A}$   $\frac{1000W}{10\times10^{-4}m^2}=\underline{1.0\times10^6\ W/m^2}$

b) $I_2=I_1\left(\frac{r_1}{r_2}\right)^2$

$I_2=1.0\times10^6\cdot\left(\frac{1}{2}\right)^2=\underline{2.5\times10^5 W/m^2}$

4a) Prove that $\theta_3=\theta_1$ if $n_3=n_1$




$n_1\sin\theta_1=n_2\sin\theta_2$

$n_2\sin\theta_2=n_3\sin\theta_3$

$n_1\sin\theta=n_2\sin_2\theta=n_3\sin\theta_3$

<u>If $n_1=n_3$, $\sin\theta=\sin\theta_3$ so $\theta_1=\theta_3$</u>

b) $M=\frac{d_i}{d_o}$   $\frac{1}{f}=\frac{1}{d_o}+\frac{1}{d_i}$

$\frac{1}{f}=\frac{1}{12}$   $2\cdot\left(\frac{1}{12}\right)-\left(\frac{1}{8}\right)\cdot3=\frac{2}{24}-\frac{3}{24}=-24cm \leftarrow d_i$

$d=8,\frac{1}{8}$   $M=\frac{-24}{8}=\underline{3.0}$

5) 200 barns at 10keV, 1cm thick

$I=I_0e^{-ux}$   density $=n=\frac{\rho N_A}{A}$   $n=\frac{11.340kg/m^3(6.022\times10^{23}mol^{-1})}{0.2072\ kg/mol}=3.295\times10^{22}$ atoms/cm$^3$

$x=1cm$   $\rho=11.34g/cm^3$

$\theta=200$ barns   $N_A=6.022\times10^{23}mol^{-1}$

$A=207.2g/mol$

$e^{3.295\times10^{22}\cdot200\times10^{-24}}=0.00137$, <u>so 0.137% of Xrays pass through</u>

6a) 250mJ in 60kg Adult

$Dose=\frac{E}{m}=\frac{0.250J}{60kg}\times100=\underline{0.416rad}$

b) $Dose=\frac{0.250J}{2kg}\times100=\underline{12.5\ rad}$

c) $0.125\times1=\underline{0.125\ SV}$

$1Gy=100rads$

$0.125Gy=12.5\ rads$

$RBE=1$

D) This dosage is not an immediate risk but it can cause longlasting effects as the limit is ~2-3mSV/year.